

Project Initialization and Planning Phase

Date	15 June 2025
Team ID	SWTID1749707061
Project Name	Online Payments Fraud Detection Using Machine Learning
Maximum Marks	3 Marks

Problem Statement:

Customers using online payment platforms face increasing risks of fraudulent transactions, leading to financial losses and reduced trust. Traditional rule-based fraud detection systems are often slow, inaccurate, and unable to adapt to new fraud patterns. This project aims to develop a machine learning-based fraud detection system that can analyze transaction data in real-time, accurately identify suspicious activity, reduce false positives, and improve security for both users and service providers. The goal is to enhance fraud prevention, protect customer data, and build trust in digital payment systems.



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An online payment service user	Make secure and reliable transactions online	I am worried about my financial information being stolen or misused due to fraudulent activities	Fraudulent transactions are often detected too late, causing financial losses and stress	Unsafe and hesitant to use digital payment platforms
PS-2	A financial institution or payment gateway provider	Identify and prevent fraudulent transactions in real-time	It's difficult to manually analyze the large volume of transactions and detect complex fraud patterns	Fraudsters constantly change their tactics and traditional rule-based systems are no longer sufficient	The need for a more intelligent, automated, and adaptive fraud detection system